PRODUCT USER MANUAL

For Arctic Time series of ice drift

SEAICE_ARC_SEAICE_L3_REP_OBSERVATIONS_011_010

Issue: 3.0

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RECORD TABLE

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2.9	20.06.2022	III.1 IV.4	Add automatic temporal extension	D. Briand	C. Wettre

2.10	16.06.2023	V.2	New SSMI datasets and format homogenization	D. Briand	C. Wettre
3.0	31.05.2024	III, IV	Add CFOSAT datasets	C. Prevost	C. Wettre

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GLOSSARY AND ABBREVIATIONS

TAC	Thematic Assembly Centre
SIW	Sea Ice and Wind
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
RMS	Root mean square
SDN	SeaDataNet (climatology)
PC	Production Centre
PU	Production Unit

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

This guide describes the regional sea ice product SEAICE_ARC_SEAICE_L3_REP_OBSERVATIONS_011_010 from the Copernicus Marine Service Sea Ice Thematic Assembly Centre (SI TAC), which data services are available in order to access them, and how to use the files and services.
These products are produced by IFREMER and disseminated by the Dissemination Unit at MET Norway. It is a time series of sea ice drift observation from satellites in the central Arctic area at medium resolution.

II PRODUCT DESCRIPTION

II.1 General information

Product Lines	SEAICE_ARC_SEAICE_L3_REP_OBSERVATIONS_011_010
Geographical coverage	-180°W ⍰ 180°E ; 60°N ⍰ 89.5 °N
Variables	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity
Analysis	No
Forecast	No
Available time series	From October 1999 – present (ASCAT, QUICKSCAT) From October 2002 – present (AMSR) From September 2020 – November 2022 (CFOSAT)
Temporal resolution	24hr average field
Target delivery time	Monthly (automatic update) at month - 2
Delivery mechanism	See **Erreur ! Source du renvoi introuvable.**
Horizontal resolution	0.5 degree (ASCAT, QUICKSCAT, CFOSAT) 0.25 degree (AMSR)
Number of vertical levels	1
Format	Netcdf CF1.4

Table 1: SEAICE_ARC_SEAICE_L3_REP_OBSERVATIONS_011_010 production line

II.2 Details of datasets

SEAICE_ARC_SEAICE_L3_REP_OBSERVATIONS_011_010		
DATASETS	VARIABLES AND UNIT or GEOGRAPHICAL COVERAGE	NAME OF VARIABLES IN THE NETCDF FILE or DOMAIN AREA
cmems_obs-si_arc_phy_my_drift-amsr_P2D	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity
cmems_obs-si_arc_phy_my_drift-amsr_P3D	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity
cmems_obs-si_arc_phy_my_drift-amsr_P6D	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity

CERSAT-GLO-SEAICE_3DAYS_DRIFT_QUIKSCAT_RAN-OBS_FULL_TIME_SERIE	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity
CERSAT-GLO-SEAICE_6DAYS_DRIFT_QUIKSCAT_RAN-OBS_FULL_TIME_SERIE	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity
CERSAT-GLO-SEAICE_3DAYS_DRIFT_ASCAT_RAN-OBS_FULL_TIME_SERIE	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity
CERSAT-GLO-SEAICE_6DAYS_DRIFT_ASCAT_RAN-OBS_FULL_TIME_SERIE	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity

CERSAT-GLO-SEAICE_3DAYS_DRIFT_QUIKSCAT_SSMI_MERGED_RAN-OBS_FULL_TIME_SERIE	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity
CERSAT-GLO-SEAICE_30DAYS_DRIFT_QUIKSCAT_SSMI_MERGED_RAN-OBS_FULL_TIME_SERIE	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity
CERSAT-GLO-SEAICE_3DAYS_DRIFT_ASCAT_SSMI_MERGED_RAN-OBS_FULL_TIME_SERIE	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity
CERSAT-GLO-SEAICE_30DAYS_DRIFT_ASCAT_SSMI_MERGED_RAN-OBS_FULL_TIME_SERIE	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity

cmems_obs-si_arc_phy_my_drift-ssmi_P3D	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity
cmems_obs-si_arc_phy_my_drift-ssmi_P30D	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity
cmems_obs-si_arc_phy_my_drift-cfosat_P3D	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity
cmems_obs-si_arc_phy_my_drift-cfosat_P6D	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity

cmems_obs-si_arc_phy_my_drift-cfosat_ssmi_merged_P3D	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity
cmems_obs-si_arc_phy_my_drift-cfosat_ssmi_merged_P30D	Depth (m) Latitude (degree) Longitude (degree) Eastward_sea_ice_velocity (m/s) Northward_sea_ice_velocity (m/s)	Depth Latitude Longitude Eastward_sea_ice_velocity Northward_sea_ice_velocity

III FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV ADDITIONAL INFORMATION ABOUT PRODUCT

IV.1 File Nomenclature

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

The nomenclature used for the SEAICE_ARC_SEAICE_L3_REP_OBSERVATIONS_011_010 products is:

<date1>-<date2>_<freqflag><avgflag>-<producer>-<level>-<parameter>-<config>-<region>-<creationdate>-<version>.nc

with

date1= start date

date2=end date

DATE:YYYMMDD

freqflag = d for daily (2d, 3d or 6d) (AMSR) / (3d, 6d or 30d) (ASCAT, QUICKSCAT, CFOSAT)

avgflag= m for mean

producer=producer

level = product level (4 for Merging)

parameter= ICEM

config=sensor_drift

region=ARC for Arctic

creationdate= creation date of the file

version=product version

Could be compressed with bzip2, file extension is then .nc.bz2

IV.2 Grid

SEAICE_ARC_SEAICE_L3_REP_OBSERVATIONS_011_010 product is made to have regular latitude and longitude grids. The dataset parameters are centered in the box of this grid.

IV.3 Domain coverage

The spatial coverage of the SEAICE_ARC_SEAICE_L3_REP_OBSERVATIONS_011_010 product is suitable for the Arctic sea ice area observations. Here is an example of the coverage. Grids type is the stereographic projection

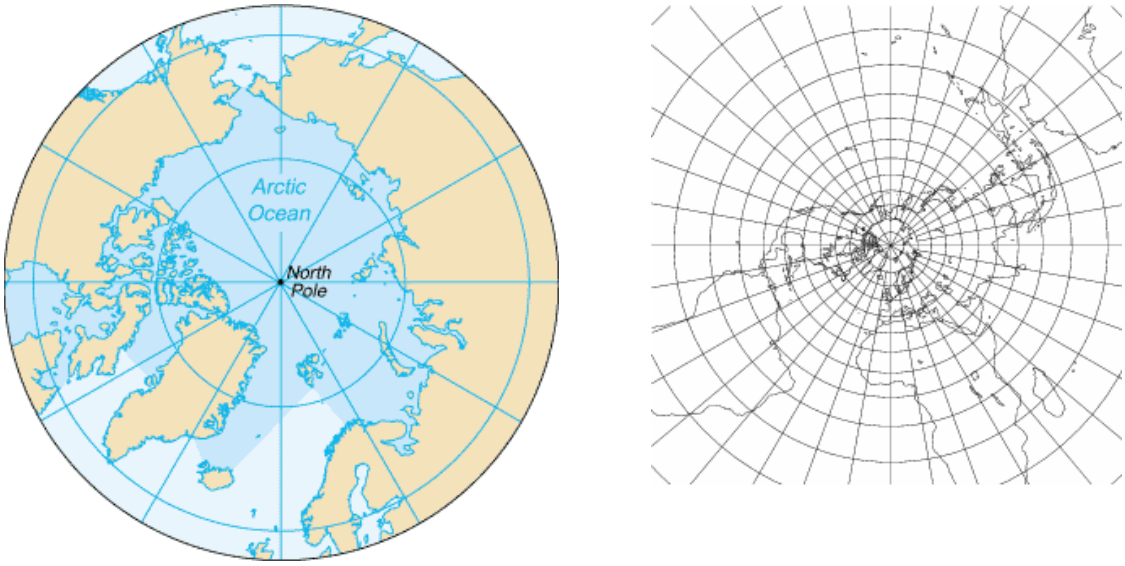

Figure 1: Spatial coverage and projection of the IFREMER SEAICE_ARC_SEAICE_L3_REP_OBSERVATIONS_011_010 product

IV.4 Update Time

SEAICE_ARC_SEAICE_L3_REP_OBSERVATIONS_011_010 products: the fields are automatically updated monthly for those which are made from data using satellites that are still in operation i.e. :

- The products are updated monthly (at the beginning of each month)

cmems_obs-si_arc_phy_my_drift-amsr_P2D
cmems_obs-si_arc_phy_my_drift-amsr_P3D
cmems_obs-si_arc_phy_my_drift-amsr_P6D
cmems_obs-si_arc_phy_my_drift-ssmi_P3D

cmems_obs-si_arc_phy_my_drift-ssmi_P30D

- The ASCAT products are updated monthly (at the beginning of each month)

CERSAT-GLO-SEAICE_30DAYS_DRIFT_ASCAT_SSMI_MERGED_RAN-OBS_FULL_TIME_SERIE
CERSAT-GLO-SEAICE_3DAYS_DRIFT_ASCAT_RAN-OBS_FULL_TIME_SERIE
CERSAT-GLO-SEAICE_3DAYS_DRIFT_ASCAT_SSMI_MERGED_RAN-OBS_FULL_TIME_SERIE
CERSAT-GLO-SEAICE_6DAYS_DRIFT_ASCAT_RAN-OBS_FULL_TIME_SERIE

- The QuikSCAT products are from 1999 until 2009 and will not be updated

CERSAT-GLO-SEAICE_30DAYS_DRIFT_QUICKSCAT_SSMI_MERGED_RAN-OBS_FULL_TIME_SERIE
CERSAT-GLO-SEAICE_3DAYS_DRIFT_QUICKSCAT_RAN-OBS_FULL_TIME_SERIE
CERSAT-GLO-SEAICE_3DAYS_DRIFT_QUICKSCAT_SSMI_MERGED_RAN-OBS_FULL_TIME_SERIE
CERSAT-GLO-SEAICE_6DAYS_DRIFT_QUICKSCAT_RAN-OBS_FULL_TIME_SERIE

- The CFOSAT product datasets are covering the 2020-2022 period. They are not updated.

IV.5 Temporal extension of analysis and forecast stored on delivery mechanism

SEAICE_ARC_SEAICE_L3_REP_OBSERVATIONS_011_010 products were made from September until May for ASCAT and QUICKSCAT. From July 2020, the production will cover the whole year, with a monthly update (for month M-1 or M-2). An exception concerns the 30DAYS DRIFT dataset which is not produced during the summer period and remains updated for the September to May period.
With regard to the AMSR datasets, the products were made from October until April. From July 2020, the production willbe updated monthly(for month M-2).

IV.6 Other information: mean centre of product, land mask value, missing value

- AMSR datasets :

The 2 day-lag products are an estimation of the mean drift between day 1 and day 3, which means that the mean velocity field is estimated with a center at 00:00 a.m on day 2.

The 3 day-lag products are an estimation of the mean drift between day 1 and day 4, which means that the mean velocity field is estimated with a center at 00:00 a.m on day 3.

The 6 day-lag products are an estimation of the mean drift between day 1 and day 7, which means that the mean velocity field is estimated with a center at 00:00 a.m on day 4.

- ASCAT/QUICKSCAT datasets :

The 3 day-lag products are an estimation of the mean drift between day 1 and day 4, which means that the mean velocity field is estimated with a center at 00:00 a.m on day 3.

The 6 day-lag products are an estimation of the mean drift between day 1 and day 7, which means that the mean velocity field is estimated with a center at 00:00 a.m on day 4.

The 30 day-lag products are an estimation of the mean drift between day 1 and day 30, which means that the mean velocity field is estimated with a center at 00:00 a.m on day 16.

- SSMI datasets :

The 3 day-lag products are an estimation of the mean drift between day 1 and day 4, which means that the mean velocity field is estimated with a center at 00:00 a.m on day 3.

The 30 day-lag products are an estimation of the mean drift between day 1 and day 30, which means that the mean velocity field is estimated with a center at 00:00 a.m on day 16.

- CFOSAT datasets :

The 3 day-lag products are an estimation of the mean drift between day 1 and day 4, which means that the mean velocity field is estimated with a center at 00:00 a.m on day 3.

The 6 day-lag products are an estimation of the mean drift between day 1 and day 7, which means that the mean velocity field is estimated with a center at 00:00 a.m on day 4.

The 30 day-lag products are an estimation of the mean drift between day 1 and day 30, which means that the mean velocity field is estimated with a center at 00:00 a.m on day 16.

V VALIDATION AND ASSESSMENT/ QUALITY CONTROL DONE ON PRODUCTS

The product has been validated over one winter. The description of the validation can be found at https://cersat.ifremer.fr/

VI ANNEX

Structure of files

```
dimensions:
        y = 720 ;
        x = 60 ;
        time = 1 ;
variables:
        int64 time(time) ;
          time:_FillValue = -9223372036854775808LL ;
          time:axis = "T" ;
          time:standard_name = "time" ;
          time:units = "days since 1980-01-01 00:00:00" ;
          time:calendar = "proleptic_gregorian" ;
          time:long_name = "reference time of field" ;
```

```
    time:authority = "CF-1.7" ;
    time:coverage_content_type = "coordinate" ;
    time:bounds = "time_bnds" ;

  double time_bnds(time, nv) ;
    time_bnds:units = "seconds since 1980-01-01 00:00:00" ;

  double lat(y,x) ;
    lat:long_name = "latitude of the start location of the displacement vector" ;
    lat:standard_name = "latitude" ;
    lat:units = "degrees_north" ;
    lat:_FillValue = 1.e+20 ;
    lat:authority = "CF-1.7" ;
    lat:valid_range = -39., -89.9. ;
    lat:coverage_content_type = "coordinate" ;
    lat:authority = "CF-1.7" ;
  double lon(y,x) ;
```

lon:long_name = "longitude of the start location of the displacement vector" ;

lon:standard_name = "longitude" ;

lon:units = "degrees_east" ;

lon:_FillValue = 1.e+20 ;

lon:units = "degrees_east" ;

lon:authority = "CF-1.7" ;

lon:valid_range = -179.7., 180. ;

lon:coverage_content_type = "coordinate" ;

double (eastward_sea_ice_velocity(time, y, x) ;

eastward_sea_ice_velocity:long_name = "eastward_sea_ice_velocity" ;

eastward_sea_ice_velocity:units = "m/s" ;

eastward_sea_ice_velocity:valid_min = -0.3858 ;

eastward_sea_ice_velocity:valid_max = 0.3858 ;

eastward_sea_ice_velocity:_FillValue = 1.e+20 ;

eastward_sea_ice_velocity:standard_name = "eastward_sea_ice_velocity" ;

eastward_sea_ice_velocity:authority = "CF-1.7" ;

```
        double (northward_sea_ice_velocity(time, y, x) ;
          northward_sea_ice_velocity:long_name = "northward sea ice velocity" ;
          northward_sea_ice_velocity:units = "m/s" ;
          northward_sea_ice_velocity:valid_min = -0.3858 ;
          northward_sea_ice_velocity:valid_max = 0.3858 ;
          northward_sea_ice_velocity:_FillValue = 1.e+20 ;
          northward_sea_ice_velocity:standard_name = "eastward_sea_ice_velocity" ;
          northward_sea_ice_velocity:authority = "CF-1.7" ;
global attributes:
        :Conventions = "CF-1.7, ACDD-1.3, ISO 8601" ;
        :Metadata_Conventions = "Climate and Forecast (CF) 1.7, Attribute Convention for Data
        Discovery (ACDD) 1.3" ;
        :_NCProperties = "version=2,netcdf=4.7.4,hdf5=1.12.0," ;
        :standard_name_vocabulary = "Climate and Forecast (CF) Standard Name Table v79" ;
        :naming_authority = "fr.ifremer.cersat" ;
        :netcdf_version_id = "4.7.4 of Oct 31 2021 03:14:43 $" ;
        :title = "Arctic Ocean - Sea Ice – 2 days Drift, AMSR" ;
        :id = "CERSAT-L3-ANT-PSIDRIFT-AMSR-2D-031" ;
```

:cmems_product_id = "SEAICE_ARC_SEAICE_L3_REP_OBSERVATIONS_011_010" ;

:summary = "This dataset provide a times series of daily sea ice drift over antarctic from AMSR sensor, every 24 hours. It is produced for the Copernicus Marine Service." ;

:references = "Sea Ice drift in the central Arctic using the 89 GHz brightness temperature of the advanced microwave scanning radiometer. Robert Ezraty, Fanny Girard-Ardhuin and Denis Croizé-Fillon, 2007" ;

:metadata_link = "None" ;

:doi = "None" ;

:processing_level = "L3" ;

:time_coverage_resolution = "P1D" ;

:time_coverage_duration = "P3D" ;

:temporal_resolution = "1 day" ;

:key_variables = "northward_sea_ice_velocity, eastward_sea_ice_velocity" ;

:keywords = "Oceans > Sea Ice > Sea ice dynamics" ;

:keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science Keywords" ;

:project = "Copernicus Marine Service" ;

:program = "Copernicus" ;

:institution = "Institut Francais de Recherche pour l\'Exploitation de la mer / Centre d\'Exploitation et de Recherche Satellitaire" ;

:institution_abbreviation = "Ifremer/CERSAT" ;

:license = "These data are available free of charge under the CMEMS data policy, refer to http://marine.copernicus.eu/services-portfolio/service-commitments-and-licence/" ;

:acknowledgment = "The development of this dataset was funded by Copernicus Marine Service" ;

:citation = ""Ifremer / CERSAT. 2022. Arctic Sea-ice Drift velocity Vectors from AMSR2 Gridded Level 4 Daily dataset. Ver. 1.0. Ifremer, Plouzane, France. Dataset accessed [YYYY-MM-DD]." ;" ;

:contact = "fanny.ardhuin@ifremer.fr;jfpiolle@ifremer.fr" ;

:technical_support_contact = "cersat@ifremer.fr" ;

:scientific_support_contact = "fanny.ardhuin@ifremer.fr" ;

:creator_email = "fanny.ardhuin@ifremer.fr" ;

:creator_type = "person" ;

:creator_institution = "Ifremer / CERSAT" ;

:creator_name = "Fanny Ardhuin" ;

:creator_url = "http://cersat.ifremer.fr" ;

:publisher_institution = "Ifremer / CERSAT" ;

:publisher_name = "Ifremer / CERSAT" ;

:publisher_url = "https://cersat.ifremer.fr" ;

:processing_software = "" ;

:source = "" ;

:source_version = "" ;

:format_version = "2.0" ;

:product_version = "1.0" ;

:doi = copernicus marine doi ;

:key_variables = "eastward_sea_ice_velocity,northward_sea_ice_velocity" ;

:platform = "GCOM" ;

:platform_type = "Sun-synchronous" ;

:platform_vocabulary = "CEOS mission table" ;

:instrument = "amsr2" ;

:instrument_type = "Imaging multi-spectral MW radiometers" ;

:instrument_vocabulary = "CEOS instrument table" ;

:polarization = "horizontal, vertical" ;

:geospatial_bounds = "POLYGON ((-180.0 -80.0, 180.0 -80.0, 180.0 80.0, -180.0 80.0, -180.0 -80.0))" ;

:geospatial_bounds_crs = "EPSG:4326" ;

:geospatial_bounds_vertical_crs = "EPSG:5831" ;

:geospatial_lat_max = 80. ;

:geospatial_lat_min = -80. ;

:geospatial_lat_resolution = 0.1 ;

:geospatial_lat_units = "degrees_north" ;

:geospatial_lon_max = 180. ;

:geospatial_lon_min = -180. ;

:geospatial_lon_resolution = 0.1 ;

:geospatial_lon_units = "degrees_east" ;

:geospatial_vertical_min = 0. ;

:geospatial_vertical_max = 0. ;

:time_coverage_start = "2022-04-20T12:00:00" ;

:time_coverage_end = "2022-04-21T12:00:00" ;

:spatial_resolution = "0.1 degree" ;

:geospatial_vertical_units = "meters above mean sea level" ;

:date_created = "2023-06-15T08:30:41Z" ;

:date_modified = "2023-06-15T08:30:41Z" ;

:cdm_data_type = "grid" ;

}